

Solutions to Exercises of Mathematical Preliminaries

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Exercise 1. *Prove that $||a| - |b|| \leq |a - b|$ (Hint: The unremarkable identity $a = a - b + b$ may be useful.)*

(Solution) We check both sides.

- $(\Rightarrow)$

$$\begin{aligned}|a| &= |a - b + b| \\ &\leq |a - b| + |b|\end{aligned}$$

by triangular inequality. Thus,

$$\Rightarrow |a| - |b| \leq |a - b|$$

- $(\Leftarrow)$

$$\begin{aligned}|b| &= |b - a + a| \\ &\leq |b - a| + |a|\end{aligned}$$

by triangular inequality. Thus,

$$\begin{aligned}\Rightarrow |b| - |a| &\leq |b - a| = |a - b| \\ \Rightarrow -(|a| - |b|) &\leq |a - b|\end{aligned}$$

Combining the two results, we have

$$||a| - |b|| \leq |a - b|.$$

□

*Comments and corrections are welcome and may be sent to hchu@nd.edu.

Exercise 2. Show that, for an arbitrary function $g : \mathbb{R} \rightarrow \mathbb{R}$, it is always true that

$$g(A \cap B) \subseteq g(A) \cap g(B)$$

for all sets $A, B \subseteq \mathbb{R}$.

(Solution) Take $y \in g(A \cap B)$, i.e. $\exists x \in A \cap B$ such that $g(x) = y$. Since $x \in A \cap B$, $x \in A$ and $x \in B$ so that $g(x) = y \in g(A)$ and $g(x) = y \in g(B)$. Thus, $y \in g(A) \cap g(B)$. $\square$

Exercise 3. Given sets A and B , define

$$A + B = \{a + b : a \in A, b \in B\}$$

Prove that if A and B are nonempty and bounded above then

$$\sup(A + B) = \sup A + \sup B$$

(Solution) 1. **WTS:** $\sup(A + B) \leq \sup A + \sup B$

Let $c = a + b \in A + B$. Then

$$a + b \leq \sup A + b \leq \sup A + \sup B.$$

Thus, $\sup A + \sup B$ is an upper bound of $A + B$. Since $\sup A + \sup B$ is the upper bound of $A + B$, it is larger or at least the same as the least upper bound of $A + B$, i.e. $\sup(A + B)$. Hence,

$$\sup(A + B) \leq \sup A + \sup B.$$

2. **WTS:** $\sup A + \sup B \leq \sup(A + B)$

Now take an arbitrary $\varepsilon > 0$. Then $\exists a \in A$ such that

$$\sup A - \frac{\varepsilon}{2} < a.$$

Also, $\exists b \in B$ such that

$$\sup B - \frac{\varepsilon}{2} < b.$$

Thus,

$$\sup A + \sup B - \varepsilon < a + b$$

Therefore,

$$\begin{aligned} \sup A + \sup B - \varepsilon &< \sup(A + B) \\ \Rightarrow \sup A + \sup B &\leq \sup(A + B). \end{aligned}$$

Given the results of 1 and 2, we have

$$\sup A + \sup B = \sup (A + B).$$

□

Exercise 4. If $\sup A < \sup B$, show that there exists an element $b \in B$ that is an upper bound for A .

(Solution) Since $\sup A < \sup B$, let

$$\varepsilon = \sup B - \sup A > 0.$$

By definition of supremum, $\exists b \in B$ such that

$$\sup B - \varepsilon < b$$

Then we have

$$\begin{aligned} \sup B - \varepsilon &= \sup B - \sup B + \sup A \\ &= \sup A \end{aligned}$$

Combining the two results, we therefore get

$$\sup A < b.$$

□

Exercise 5. Compute, without proofs, the suprema and infima (if they exist) of the following sets:

$$1. A \equiv \left\{ \frac{m}{n} : m, n \in \mathbb{N} \text{ with } m < n \right\}$$

$$2. B \equiv \left\{ \frac{(-1)^m}{n} : m, n \in \mathbb{N} \right\}$$

$$3. C \equiv \left\{ \frac{n}{3n+1} : n \in \mathbb{N} \right\}$$

(Solution) 1. $\sup(A) = 1$ and $\inf(A) = 0$

since

- if $m = 1$ and $n \rightarrow \infty$, then $\frac{1}{n} \rightarrow 0$.

- if $m = n - 1$ and $n \rightarrow \infty$, then $\frac{n-1}{n} = 1 - \frac{1}{n} \rightarrow 1$.

$$2. \sup(B) = 1 \text{ and } \inf(B) = -1.$$

since

- if $n = 1$ and m even, then 1.
- if $n = 1$ and m odd, then -1 .

3. $\sup(C) = \frac{1}{3}$ and $\inf(C) = \frac{1}{4}$.

since

- if $n = 1$, then $\frac{1}{4}$.
- if $n \rightarrow \infty$, then $\frac{1}{3}$.

□

Exercise 6. Let $a < b$ be real numbers and consider the set $T = \mathbb{Q} \cap [a, b]$. Show $\sup T = b$.

(Solution) First, we show that b is an upper bound for T . Take $t \in T$ which implies that

$$t \in \mathbb{Q} \quad \text{and} \quad t \in [a, b]$$

In particular, $t \leq b \forall t \in T$. Thus, by the definition of the upper bound, b is an upper bound of T .

Next, we show that b is the least upper bound of T . Take an arbitrary $c \in \mathbb{R}$ so that $c < b$. We know that $a < b$ and $c < b$, leading to

$$\max\{a, c\} < b.$$

Since a and c are both real numbers and by the density of $\mathbb{Q}$ in $\mathbb{R}$, $\exists q \in \mathbb{Q}$ such that

$$\max\{a, c\} < q < b.$$

In particular,

$$a < q < b \iff q \in [a, b].$$

Since $q \in \mathbb{Q}$, it follows that $q \in \mathbb{Q} \cap [a, b] = T$. We also have $c < q$, thus c is not an upper bound of T . Therefore, b is an upper bound of T , and every real number smaller than b fails to be an upper bound, i.e. b is the least upper bound. Therefore,

$$\sup T = b.$$

□